

Spinout Engine engineers ventures from raw technical research

The Core Loop

Transforms raw technical papers into structured startup theses.

The Engine

Powered by 5 parallel AI agents seamlessly orchestrated via a high-performance stack.

The Output

Generates a comprehensive venture memo automatically.

The Stress Test

Simulates a live investor room featuring dynamic questions, synthetic voice, and critical feedback.

Spinout Engine

Turn research papers into venture-ready memos, then stress-test the pitch with a synthetic AI investor.

Upload paper 

Try demo paper 



 Static HTML

 FastAPI

 Scaleway

 ElevenLabs

 Firebase

 Codex

Founders build from structured intelligence instead of blank pages



1. Drop (Input)

Users securely upload research via PDF, DOCX, TXT, or MD (up to 20MB payload limit).

2. Analyze (Extraction)

The backend automatically extracts technical novelty, market vectors, and hidden risks from the raw text.

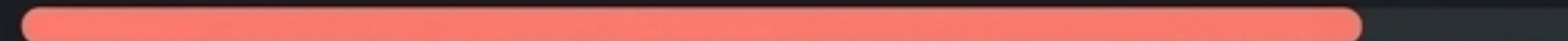
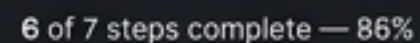
3. Pitch (Output)

The system delivers a fully structured venture memo, investor milestones, and an improved pitch narrative.

Real-time backend execution dictates the frontend pipeline

7 Phases. 5 Agents. Zero Fake Loading.

- **Strict Synchronization:** The UI strictly waits for actual JSON payloads from the server before advancing—guaranteeing that the demo never finishes before the data is ready.
- **Visible Pipeline:** Users watch the processing sequence live: Ingesting Document → Extracting Technical Novelty → Mapping Market Wedge → Checking Competitors → Identifying Investor Risks → Building Venture Memo → Preparing Investor Room.



Paper Intake Agent :

Complete

Technical Novelty Agent

Complete

Market Wedge
Agent

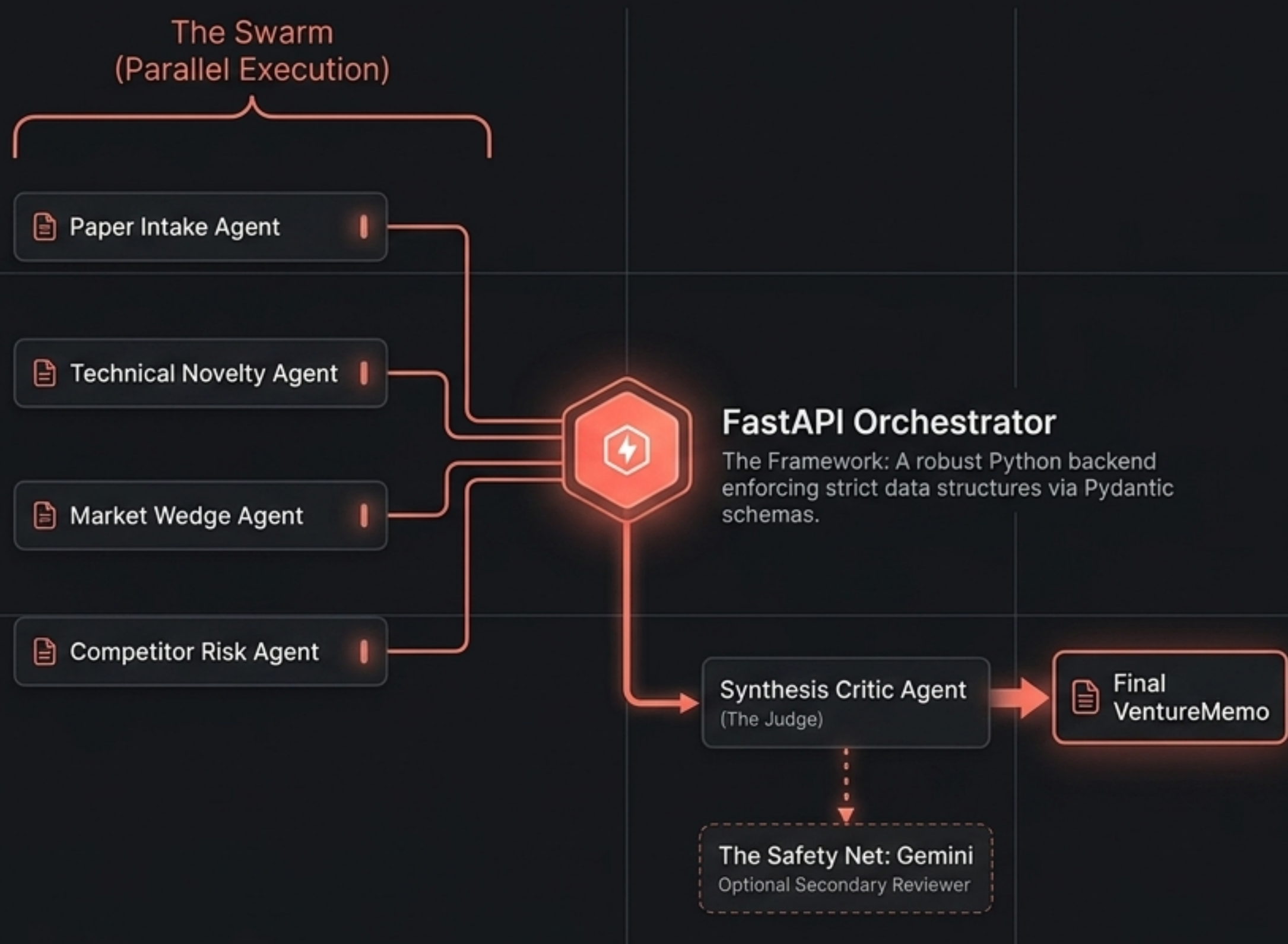
Complete

Preparing Investor Room

Running

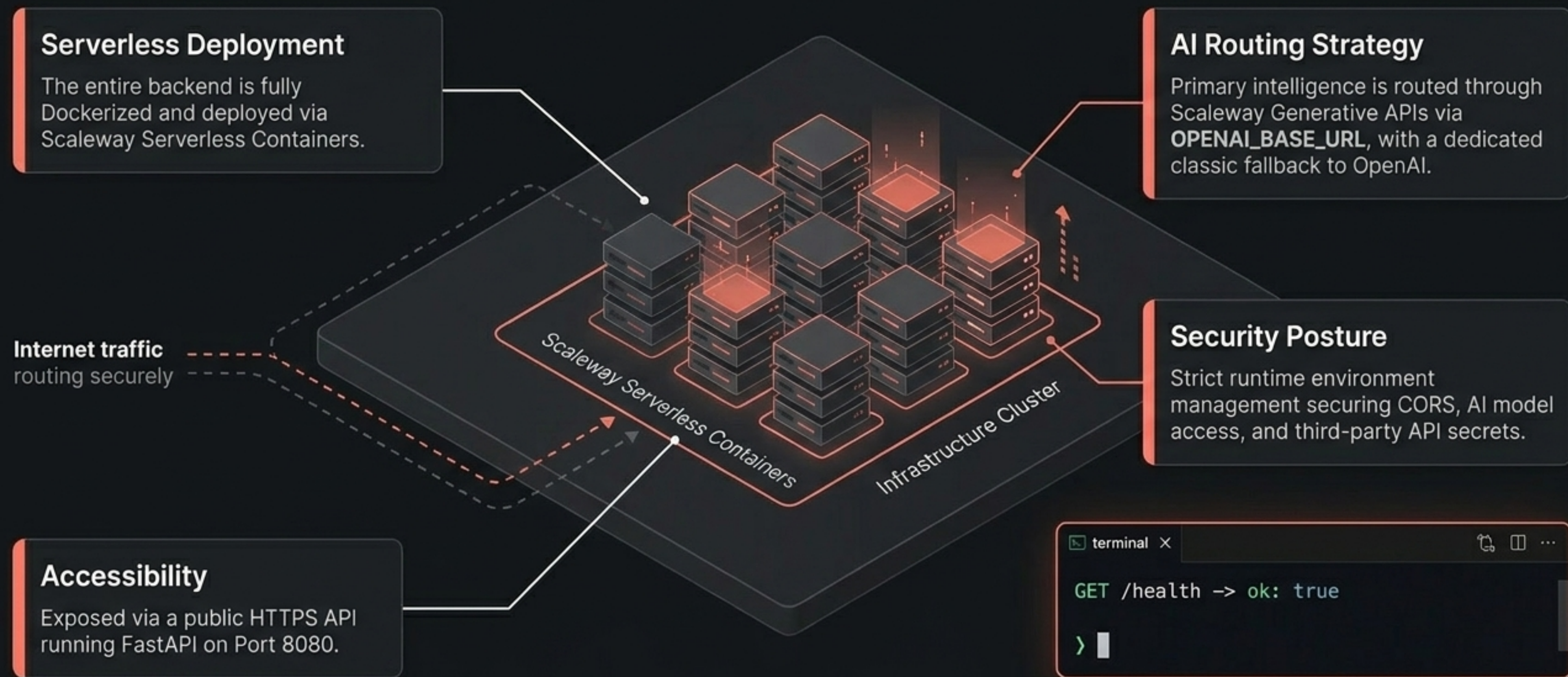
100

Pydantic schemas and FastAPI orchestrate the agent swarm

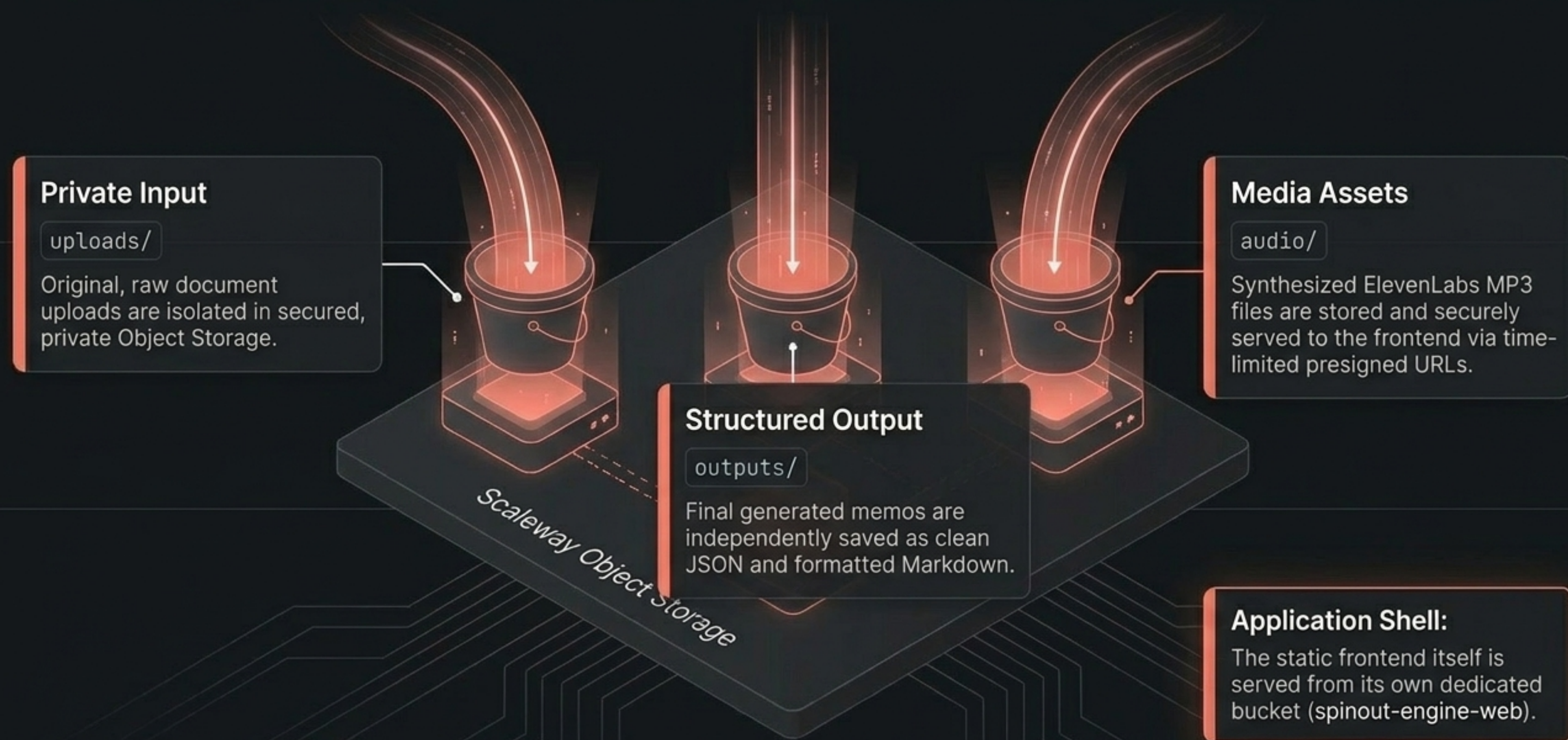


```
orchestrator.py x
orchestrator.py
1 from fastapi import FastAPI, BackgroundTasks
2 from pydantic import BaseModel
3 from typing import List
4 import asyncio
5
6 # Define Pydantic Schemas for structured data
7 class AgentRequest(BaseModel):
8     document_id: str
9     payload: dict
10
11 class AgentResponse(BaseModel):
12     agent_name: str
13     analysis: dict
14
15 app = FastAPI()
16
17 async def run_agent_swarm(doc_id: str):
18     # Execute agent swarm in parallel using asyncio
19     tasks = [
20         process_with_agent("paper_intake", doc_id),
21         process_with_agent("technical_novelty", doc_id),
22         process_with_agent("market_wedge", doc_id),
23         process_with_agent("competitor_risk", doc_id),
24     ]
25     results: List[AgentResponse] = await asyncio.gather(*tasks)
26
27     # Route to Synthesis Critic Agent for final memo
28     final_memo = await synthesis_critic.analyze(results)
29     # Optional safety check
30     if safety_config.enable_gemini:
31         await gemini_safety_net.review(final_memo)
32
33     return final_memo
34
35 @app.post("/orchestrate/venture_memo")
36 async def create_venture_memo(request: AgentRequest, background_tasks: BackgroundTasks):
37     # Trigger orchestrator in background
38     background_tasks.add_task(run_agent_swarm, request.document_id)
39     return {"status": "Swarm Orchestration Initiated", "doc_id": request.document_id}
```

Scaleway Serverless powers the core execution runtime



S3-compatible architecture isolates distinct data streams



Synthetic VC voices interrogate the finalized pitch

Dynamic Generation

The dedicated `/investor/question` endpoint spawns a synthetic, skeptical Deep-Tech VC persona.

Voice Synthesis

ElevenLabs dynamically converts the AI's challenge into high-fidelity MP3 audio.

Demo-Proof Resilience (Triple-Tier Fallback)

- Primary:** Audio saved to Object Storage and played via S3 Presigned URL.
- Secondary:** If S3 fails, the system falls back to a direct Audio Base64 stream.
- Tertiary:** If ElevenLabs fails entirely, it defaults to a text-only output to guarantee the demo never crashes.

The screenshot shows a user interface for a synthetic VC persona named Mara, a "Skeptical Deep-Tech VC". The interface includes a status bar with "Synthetic voice ready" and "Transcript ready". It displays "QUESTION 1 / 4" and a question from Mara: "What is the quantified cost per validated high-affinity binder on your platform versus the current Rosetta workflow, and how does that translate into a concrete willingness-to-pay from biotech customers?". Below the question is a "Voice Transmission" section with a waveform and "LIVE" status, featuring "Play audio" and "Stop" buttons. At the bottom is an "Ask next question" button with a right arrow. On the right side, there is an "INVESTOR HINT" section with the text "Lead with willingness-to-pay evidence, downtime economics, and the", a "YOUR ANSWER" section with a text input field labeled "Type your answer as the founder...", and a "Submit answer" button. A large audio waveform is visible at the bottom right of the interface.

A lightweight vanilla frontend relies on hardened Firebase auth

Zero-Bloat Stack

Built with vanilla HTML/CSS/JS with absolutely no bundler bloat.

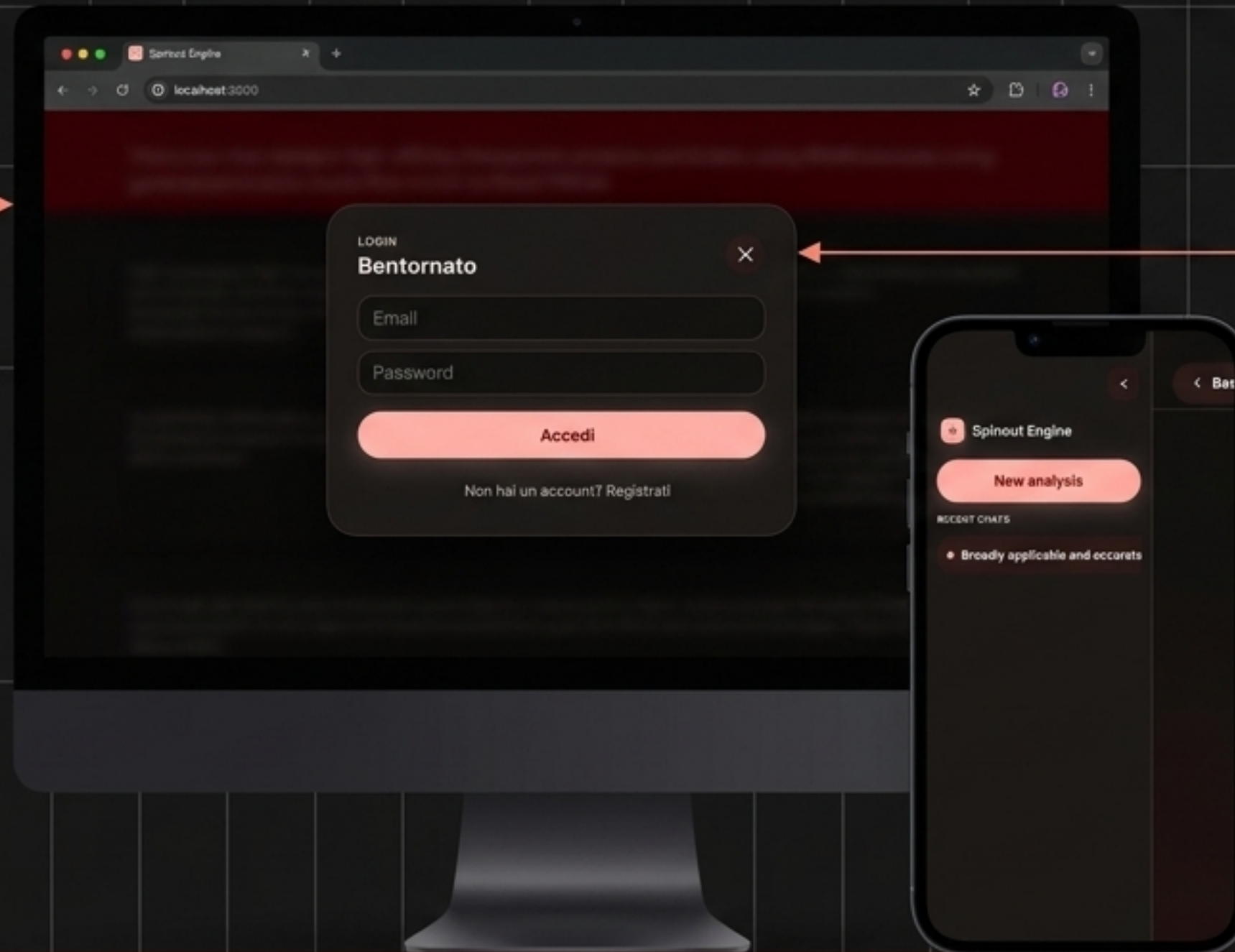
Runtime configuration is handled cleanly via `env.js`.

Identity Management

Integrated Firebase Auth via SDK CDN managing registration, secure login, logout, and automatic session restoration.

Persistent UX

Features a responsive, ChatGPT-style collapsible sidebar that allows logged-in users to easily track and resume Recent Analyses.



Codex operated as an active engineering copilot

Design to Reality

Translated raw UI prototypes directly into a functional, live FastAPI/Vanilla application.

a7b8f9d

```
commit a7b8f9d · connect  
frontend auth to  
firebase
```

e3c4d5f

```
commit e3c4d5f · sync  
processing progress with  
backend result
```

Infrastructure as Code

Authored the critical Dockerfiles, Dockerfiles, Scaleway configurations, parsed document setups, and strict CORS policies.

Active Debugging

Resolved complex friction points including WSL/Docker networking, ElevenLabs audio streaming, frontend progress syncing, and Firebase state management.

g6h7i8j

```
commit g6h7i8j · add auth  
shell and recent  
analyses sidebar
```

The entire ecosystem is deployed, functional, and ready for scale

1 Live End-to-End Execution

- Real document uploads are successfully parsing.
- Structured AI Memos are generating in real-time.
- The ElevenLabs synthetic voice is actively stress-testing users.

2 Infrastructure Verified

Both Scaleway serverless containers and Firebase Authentication are fully integrated and handling live traffic.

3 Tangible Output

Users instantly export actionable data via Markdown and JSON payloads.

Pitch Improvement Report Investor test complete

ORIGINAL PITCH

"We're building the first AI platform that turns a single molecular specification into a high-affinity therapeutic peptide in weeks, not months. By fine-tuning RoseTTAFold into a diffusion model RFDiffusion, we achieve experimental success rates two orders of magnitude higher than Rosetta, hitting experimental identity picomolar binders by testing only dozens of candidates. Our Back solution slashes design costs, accelerates drug discovery, and opens new opportunities for genome and metabolite-targeted engineering."

IMPROVED PITCH

"AI-aliven platform that designs high-affinity therapeutic proteins and binders using RFDiffusion, a diffusion-based generative model fine-tuned on RoseTTAFold. The first wedge is Rapid design of picomolar affinity peptide binders to specific therapeutic wedge to g., Desirable helical peptides) by generating a small, high-quality candidate set with RFDiffusion. The urgent pain is De novo protein binder design today requires synthesizing and testing tons of thousands of candidates, resulting in high material

30 / 60 / 90 DAY RECOMMENDATIONS

- 1-30 DAYS** Feature and model software pipeline (design generation → AlphaFold2 validation).
- 30-60 DAYS** Set up cloud infrastructure and user authentication.
- 60-90 DAYS** Launch beta program with 2-3 selected clinical partners, deliver first set of peptide binder designs.
- 90-120 DAYS** Publish benchmark comparing RFDiffusion hit rates to Rosetta on a public dataset.
- 120-150 DAYS** Secure first paying enterprise customer (subscription + design fee).
- 150-180 DAYS** Release API documentation and developer portal.

KEY RISKS

- Model** Experimental validation cost may remain high if success rates do not scale beyond benchmark targets. Partner with contract research organizations to accelerate early testing and model refinement in collaboration to improve hit quality.
- Market** Betascope on a fine-tuned RoseTTAFold could cause IP infringement issues. Negotiate clear licensing terms with the RoseTTAFold developers; develop an independent naming pipeline as a fallback.
- Business** Generative AI business models (subscription) may be limited. Continuously expand training data with diverse backbones, incorporate active learning loops where latest designs feed back into model retraining.
- Legal** Adoption barriers due to need for customers to have wet-lab capability. Offer an end-to-end service that includes synthesis and initial functional testing through partner labs.
- Adoption** Regulatory uncertainty for therapeutic proteins generated by AI. Engage regulatory consultants early; generate full documentation of design provenance and validation data to support IND filings.

MARKETING MESSAGES

- Value** General applicability of RFDiffusion beyond the experimentally tested targets (e.g., diverse enzyme classes).
- Use** Consistent picomolar affinity across unrelated therapeutic targets.
- Cost** Quantified cost savings for customers benefiting from willingness to pay.

INVESTOR READINESS

70

out of 100

Needs proof points

BENCHMARK CLAIMS

- RFDiffusion hit rates vs. current Rosetta pipelines 70%
- The operational strategy is sustainable, scalable 65%
- Small design of picomolar affinity peptides feasible 88%
- 4 source peptides 95%

TOP INVESTOR QUESTIONS

- What is the gross margin cost per successful design compared with current Rosetta pipelines?
- How will you ensure secure IP protection for designs generated by RFDiffusion?

Investor scorecard
Design
IP protection
Scalability
Provenance

EXPORT

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